Dhole Patil College of Engineering

First Year of Engineering (2024 Pattern)

Course Code: IKS-151 Course Name: Indian Knowledge System

Assignments for Term Work

Assignment 3

Students should complete the assignment as stated in the Instructions.

Learning Outcome: Develops research skills while enhancing knowledge of Indian scientific achievements.

NOTE: Enhance your presentation by including relevant pictures, diagrams, and illustrations. Make it visually appealing with decorative elements such as borders, color themes and pictures that reflect the cultural and environmental aspects of your project.

Choose any 5 topics and write 500 words paragraph for each.

Topics:

1. **Vedic Mathematics**
    a. Mental math techniques and algebraic formulas.
2. **The Concept of Zero (Brahmagupta)**
    a. How ancient Indian mathematicians revolutionized mathematics.
3. **The Decimal System & Place Value Notation**
    a. Transforming global mathematics with numerical innovations.
4. **Languages & Linguistics (Panini's Grammar)**
    a. The earliest known comprehensive grammar system for Sanskrit.
5. **Ayurveda & Early Medicine (Charaka Samhita)**
    a. Holistic health systems, plant-based medicine, and surgery.
6. **Sushruta's Surgical Techniques**
    a. The world's first documented plastic surgery and surgical tools.
7. **Astronomy & Planetary Movements (Aryabhata)**
    a. The heliocentric theory, eclipses, and Earth's rotation.
8. **Metallurgy & the Iron Pillar of Delhi**
    a. Advanced metalworking techniques that resisted corrosion.
9. **Architecture & Engineering (Stepwells, Temple Design)**
    a. Structural ingenuity, geometry, and sustainable water systems.
10. **The Science of Yoga & Meditation**
    a. Ancient practices that laid the foundation for modern well-being.

11. **Botany & Agriculture (Krishi Parashara)**
    a. Early texts on plant life, soil science, and sustainable farming.
12. **Textiles & Natural Dyes**
    a. Pioneering techniques in fabric weaving, dyeing, and embroidery.
13. **Timekeeping & Calendar Systems (Jyotisha)**
    a. Early calculations of time, seasons, and celestial cycles.
14. **Alchemy & Early Chemistry (Rasashastra)**
    a. Ancient practices of metal purification and medicinal compounds.
15. **Logical Debates & Philosophical Inquiry (Nyaya & Vaisheshika)**

The Write Up should contain:

1. **Historical Background**

- When, where, and by whom the discovery was made.

2. **Scientific Principles**

- Explanation of the concept and how it works.

3. **Overview**

- Overview of the discovery and why it matters.

4. **Impact on Ancient Society**

- How people used this knowledge in daily life or rituals.

5. **Global Influence & Modern Applications**

- How this discovery inspired future scientists or shaped modern science.

6. **Conclusion & Reflection**

- Lessons we can take from ancient Indian science.

Writing Rules:

- Paragraph form (no bullet answers)
- Simple, clear language
- No plagiarism, No copying from others (5 marks will be deducted if found so)
- Proper headings compulsory